



Shahid Singh

Systems Developer

Automation engineer, full-stack & embedded developer. I build systems that eliminate busywork — ticket workflows, browser tools, hardware controllers, and cross-platform apps — all from Newcastle, South Africa.

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Newcastle, South Africa

Experience

Systems Developer — Jenny Internet

Jun 2025 – Present · 1 yr

Newcastle, South Africa

- Architected the entire DFE (Dynamic Form Engine) ticket workflow system reducing manual ticket handling by ~70% across Support, Operations, and Admin teams.
- Built 20+ production workflow templates including Wireless Support, Customer Callout AI, Equipment Orders, Vehicle Management, Overtime, Expenses, Site Rental, and Admin requests.
- Designed guided troubleshooting flows, Quick Navigate, pre-escalation checklists, customer-in-pain detection, and cross-ticket automation pipelines.
- Integrated Rocket.Chat alerting across all departments for real-time operations visibility.
- Created internal wiki documentation, beginner guides, and wireless troubleshooting playbooks used company-wide.
- Developed Chrome extensions (Jenny Screenshot Assistant, Jenny Vault) for agent productivity with OCR, redaction, annotation, and encrypted template storage.

Senior Support Agent — Jenny Internet

Jun 2023 – May 2025 · 2 yrs

Newcastle, South Africa

- Provided advanced wireless ISP support including no-connection, speed, intermittent, and outage troubleshooting.
- Performed signal analysis, traceroutes, tower/sector checks, LAN sync verification, and queue inspection.
- Mentored junior agents and contributed to internal knowledge base and troubleshooting guides.
- Developed deep understanding of customer pain points that later informed workflow automation design as a developer.

College Lecturer & Server Administrator

2020 – 2023 · 3 yrs

Newcastle, South Africa

- Delivered lectures in IT and computer-related subjects to tertiary students.
- Managed and maintained the college server infrastructure including user accounts, backups, and network stability.
- Developed course materials and practical assessments for systems development modules.

Education

Diploma in Systems Development

Boston City Campus · Newcastle, South Africa · 2025 – Present

- Systems Development (HSYD100-1)
- Information Systems Management (HISM100-1)
- Academic Literacy (HALP100-1)

Skills & Technologies

Workflow & Automation

Heimdall Triggers SendActions DFE Templates Ticket Automation Rocket.Chat API

JSON Workflows

Frontend & Browser Extensions

React TypeScript Manifest V3 Vite HTML/CSS/JS Tesseract.js OCR

Canvas API

Mobile & Desktop Apps

Flutter / Dart Riverpod Drift / SQLite Supabase GoRouter

Embedded & Hardware

ESP32 / Arduino BLE HID LVGL TFT_eSPI C++ 3D Printing Web Serial

Backend & AI

Python FastAPI Ollama / vLLM ChromaDB RAG Docker PostgreSQL

Operating Systems & DevOps

Linux (Arch/Debian) Git systemd GitHub Actions Cross-compilation ARM64

Key Achievements

- ◆ Reduced manual ticket handling by ~70% across ISP support operations through custom workflow automation.
- ◆ Built a production-ready BLE macro keyboard (SudoDeck) competing with the Elogato Stream Deck — sold in South Africa for R1,099 with full open-source firmware and tools.
- ◆ Created a 249MB offline educational suite (NexCampus) with 16 study tools, code lab, OCR, dictionary, and interactive solar system — zero internet required.
- ◆ Developed an open-source business management suite (SpiceDesk) for small food shops with POS, inventory, CRM, and expense tracking — MIT licensed.
- ◆ Published 6+ open-source projects on GitHub including Chrome extensions, Flutter apps, and embedded firmware.
- ◆ Self-hosted AI workspace (Odysseus) supporting local LLMs, RAG, agents, email triage, and deep research.
- ◆ Built a custom Linux kernel (SudoOS) for ARM64 handheld gaming devices.

Work Projects — Jenny Internet

DFE Ticket Workflow Templates

20+ templates for Support, Operations, and Admin — Wireless Support, Callout AI, Equipment Orders, Overtime, Vehicle Management, Expenses, Site Rental, Admin Requests.

Wireless Support 02 Workflow

Guided troubleshooting for no connection, speed, intermittent, password, outages, callouts, HOD escalation, manager calls, and Advanced Support.

Quick Navigate

Jump-to-section navigation for long support tickets — Log Details, Test No Connection, Callout, Escalation, Outage, and more.

Customer Callout AI

Structured callout tickets with customer, service, dealer, priority, payment, and troubleshooting — auto-received from Wireless Support.

Callout Automation Logic

Customer-paid, Jenny-paid, dealer-paid callouts with payment, arrangement, completion, cancellation, and dealer-change follow-ups.

Heimdall Scheduled Triggers

Dealer-change triggers, duplicate prevention, cancellation handling, and automatic trigger deregistration.

HOD Escalation Workflows

Original ticket state saved, assigned to HOD/manager, returned to original state and agent after review.

Manager Call Requests

Ticket assignment to management with notification, original requester saved, auto-return on completion.

Rocket.Chat Notifications

Real-time alerts for managers, HODs, support agents, operations, callouts, overtime, and system alerts.

Customer-in-Pain Detection

Identifies customers with repeated support issues within a configurable recent period.

Overtime Logging

Technician start/end tracking, duration logging, technician assignment, and Rocket.Chat updates.

Vehicle & Expense Linked Workflows

Expense tickets update vehicle-related totals and linked records bidirectionally.

Equipment Order UX & Automation

Order item handling, billing, compact infoblocks, required fields, cleaner processing.

CPE Installation Workflows

MikroTik script handling, manual override logic, and cleaner install processing.

Admin/System Request Workflows

Custom checklists and visible task progress on tickets.

Refund & Site Rental Workflows

Suspend/restart handling, structured payment terms.

Internal Wiki & Documentation

Heimdall triggers, ticket watchers, wireless troubleshooting, beginner support processes.

Form UX Improvements

Required fields, headings, infoblocks, helper text, collapsible sections, checklists, guided steps.

Ticket-to-Ticket Automation

One workflow creates/updates another related ticket and passes structured data between them.

Chrome Extensions for Agents

Jenny Screenshot Assistant (OCR + redaction + ticket notes) and Jenny Vault (encrypted template storage).

Personal Projects

SudoDeck

C++ ESP32 BLE Arduino Python LVGL

Bluetooth macro keyboard — wireless Stream Deck competitor. ESP32+ILI9341 touchscreen, BLE HID, multi-page grid, 30 buttons/page, app launcher, live widgets, themes, Web Serial config. Sold in SA for R1,099.

[GitHub](#) [Website](#)

SpiceDesk

Flutter Dart Supabase Riverpod

Open-source business management suite for bakeries, delis, and food shops. POS, inventory, CRM, expenses, quotes, reports, WhatsApp sharing, invoice PDFs. Free forever, MIT licensed.

[GitHub](#)

Odysseus

Python FastAPI ChromaDB Ollama Docker

Self-hosted AI workspace — chat with local LLMs, agent with MCP tools, deep research, document editing, email triage, calendar, notes, tasks, memory, and 2FA. Privacy-first, runs on your own hardware.

[GitHub](#)

NexCampus

[Python](#)[JavaScript](#)[OCR](#)[PyWebView](#)

Offline educational suite — 249MB with 16 study tools, code lab (Python/JS/Bash), text tools, 262k-word dictionary, periodic table, interactive solar system, OCR, 9 themes. Free, no internet required.

[GitHub](#)

Family Tree App

[Flutter](#)[Dart](#)[SQLite](#)[Riverpod](#)

Interactive offline family tree builder. Multiple family books, tree view with pan/zoom, person profiles, photos, relationships, timeline, statistics, dark/light themes. Cross-platform (Windows, Android, Linux).

[GitHub](#)

Jenny Screenshot Assistant

[React](#)[TypeScript](#)[Vite](#)[OCR](#)[Manifest V3](#)

Production Chrome/Edge extension for screenshot capture, local OCR, regex extraction, redaction, annotation, and ticket note generation. Privacy-first — no uploads by default, optional AI diagnosis.

[GitHub](#)

Jenny Vault

[JavaScript](#)[Chrome API](#)[Encryption](#)

Chrome extension for saving template JSON/text with local encryption. Password-protected vault with search, edit, JSON/tree views, export, and bulk management.

[GitHub](#)

CYD Chrome Controller

C++

ESP32

Arduino

Python

ydotool

Physical touch interface for browser media control. ESP32 + touchscreen with 9 media zones (prev/play/next/vol/mute/back/reload/forward). Serial communication with Python host using ydotool.

[GitHub](#)

Agent Robin 007

HTML

CSS

JS

Canvas API

James Bond-themed Father's Day interactive experience. Animated mission briefing, 3-round shooting gallery, score tracking, Bond quotes, downloadable commendation card, confetti celebrations.

[GitHub](#)

For Akiva, With Love

HTML

CSS

JS

Canvas

Web Audio

Interactive web experience — a garden of 9 years of memories. Plant seeds, watch flowers bloom with love letters and photos. Aurora sky, fireflies, moonlit scene, ambient music.

[GitHub](#)

SudoOS

Linux Kernel

ARM64

Bash

Cross-compile

Custom Linux kernel build for ARM64 handheld devices (R36S). Automated build script, custom kernel config, initramfs, ulnitr creation, logo customization.

[GitHub](#)

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Built with a single HTML file · [Source on GitHub](#)